(H4) FOR THE CUP SQUARE: $[K(\mathbb{Z},2),K(B,4)]_* \cong \mathrm{Hom}(\mathbb{Z},B)$

Purpose and status. Informal proof for §2 of `doc/pi3-s2-postnikov-handoff-4.txt`:

h4 : KZ2Cohomology.H4 CupSquare.kappa,

that is, for every abelian group B the map $\Phi_B : \mathrm{Hom}(\mathbb{Z},B) \to [K_2, K(B,4)]_*$, $g \mapsto [\kappa; K(g,4)]$ is an equivalence. Written before the formal proof; the numbered statements are meant to be declarations (§8). Foundations: those of the development (univalence, images; function extensionality derived). No new axiom; the sphere axiom is not used; no pushout, no spectral sequence. *Refereed* independently (a fresh reader given the note and the library): correct with fixes, all incorporated here (explicit Lemma 9 section instead of `rt-base`; the base point of the target of T; `conn-rt-n`; the conjugation in Lemma 15; the single hypothesis $\mathsf{isTrunc}\,6\,Y$; Lemma 17 via pointing irrelevance).

The idea in one paragraph. Write $M_Y = \mathrm{Map}_*(K_2, Y)$. A bipointed map $c : K_2 \to_* M_Y$ has a diagonal $\Delta c = (x \mapsto (c\,x)(x)) : K_2 \to_* Y$, and $\kappa = \Delta c_2$ by definition. We prove two theorems:

(A) (*every class is a diagonal*) for $Y = K(B,4)$, $c \mapsto [\Delta c]$ is a bijection $\mathrm{Map}_*(K_2, M_Y) \to [K_2, Y]_*$;

(B) (*bilinear classification*) $g \mapsto c_2; K(g,4)_*$ is a bijection $\mathrm{Hom}(\mathbb{Z},B) \to \mathrm{Map}_*(K_2, M_{K(B,4)})$, where G_* is postcomposition.

Since $\Delta(c_2; K(g,4)_*) \equiv \kappa; K(g,4)$ (by definition, §6), Φ_B is the composite of the two bijections, hence an equivalence. (A) is where the information about the *diagonal* is, and no factor 2 appears: the classical content is $\Delta^*(u \otimes u) = u^2$ in $H^4(K(\mathbb{Z},2))$, not the polarisation $\mu^*(u^2) = u_1^2 + 2u_1u_2 + u_2^2$. The proof of (A) uses DW1's Lemma 9 (the Yoneda step over the based path space $(a : K_2) \times (a = *)$, i.e. the path fibration $K(\mathbb{Z},1) \to * \to K(\mathbb{Z},2)$) to embed the set of classes into a set of "multiplicativity defects", and computes the defect of a diagonal by one path induction (Proposition 10). (B) uses only deloopings, the circle property and the normalisation `cup-norm` of c_2.

The trap. No step recovers a coefficient from a cross term $c(x)(y) + c(y)(x)$. The invariant of a class F (§4) is computed on F itself; on $F = \Delta c$ it is the looping of c, so for $B = \mathbb{Z}/2$ the class $\kappa; K(1,4)$ is detected (it is $\Delta(c_2; K(1,4)_*)$, and $c_2; K(1,4)_* \neq 0$ by (B)).

1. Conventions and interfaces

As in `doc/cup-square.pdf`: `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected"; $K_m = K(\mathbb{Z},m)$, $K_m^B = K(B,m)$ (`EMMaps.KK[B].K m`), pointed by $*$; $p \cdot q$ is `trans p q` (diagrammatic, by J on q, so $p \cdot \mathsf{refl} \equiv p$); $\Omega Y = (y_0 = y_0, \mathsf{refl})$; $F; G$ is `ptcomp F G` (F first). For pointed X, Y: $\mathrm{Map}_*(X,Y) = (F : X \to Y) \times (F\,x_0 = y_0)$, pointed by $0 = (\lambda_.\, y_0, \mathsf{refl})$.

DW1 notation (`DeloopMaps.Maps[A,a0,B,b0]`, here always $A = K_2$, $a_0 = *$): $\mathsf{PMO} = \mathrm{Map}_*(\Omega A, \Omega B)$, pointed by $f_0 = (\lambda q.\mathsf{refl}, \mathsf{refl})$; for $h : (a = a_0) \to (b = b_0)$ and $p : a = a_0$, $D\,a\,b\,h\,p = (q \mapsto (h\,p)^{-1} \cdot h(p \cdot q),\ \ldots) : \mathsf{PMO}$; $\mathsf{Kf}\,F\,F_0\,a\,p : F\,a = b_0$ by J on p ($\mathsf{Kf}\,F\,F_0\,a_0\,\mathsf{refl} \equiv F_0$); $\mathsf{loop}\,F = D\,a_0\,(F a_0)\,(\mathsf{Kf}F\,a_0)\,\mathsf{refl}$; $C\,a\,b = (h : (a = a_0) \to (b = b_0)) \times \prod_{p:a=a_0}(f = D\,a\,b\,h\,p)$; $\mathsf{Unloop}\,f = (F : \mathrm{Map}_*(A,B)) \times (f = \mathsf{loop}\,F)$; $\mathsf{Rel} = \prod_a (b : B) \times C\,a\,b$.

Interfaces used (all in the library, or in §2a/§2b):

Date: 28 October 2026.

(I1) `KK`: $\mathsf{isConn}\,(m+2)\,K^B_{m+1}$, $\mathsf{isTrunc}\,(m+3)\,K^B_{m+1}$, pointed equivalences $e_m : \Omega K^B_{m+1} \simeq_* K^B_m$; `KMap[A,B,g].K m` $= K(g,m)$ with `loopK`: $\mathsf{loop}\,K(g,m+1) = e_m; K(g,m); e_m^{-1}$; `KFun`: $K(h\circ g,m) = K(g,m); K(h,m)$.
(I2) `Lemma4`: `lemma4`, `cor5-map`; `PtSec`.
(I3) `DeloopMaps`: `lemma9` (a retract $\mathsf{Unloop}\,f \lhd \mathsf{Rel}$), `Levels` (`s1, r1, s2, r2, s3, r3`, `cell`, `cor10`), `cor11`, `Mult.cor12`, `ptmap-path`, `E`.
(I4) `LoopCalc`: `loop-comp` ($\mathsf{loop}(F;G) = \mathsf{loop}F; \mathsf{loop}G$), `unique-loop`, `unique-loop1`, `ptcomp-assoc`, `pt-inv-l/r`, `comp-const-l`, `loop-const`.
(I5) `CircleProperty.Circle`: `evPt`, `pointed-equiv`, `evPt-conj`, `evPt-loop`, `ptmap-conn`.
(I6) `CupSquare`: `PtMapLoops` (`ev`, `ev-ap`, `unev`, `ev-unev`), `iota`, `e1`, `e1inv`, `c2`, `c2-loop`, `kappa`, `tM4`, `cup-norm`.
(I7) `LoopConj` (`conj`, `conj-trans`, `ap-homotopy`), `EH.Abelian` (loops of a type pointed equivalent to a loop space commute), `PiOne.loop-mult`.
(I8) `H3Proof`: `rt-ptmap`, `h3`; `Bijection.Bij`; `HomotopyClasses` (`HC`, `hcl`, `effective`, `ind`); `Connected` (`conn-id`, `conn-elim`).

2. Three small lemmas

Lemma 1 (pointing irrelevance, sets)**.** *If* $\mathrm{Map}_*(X,Z)$ *is a set and* $\Omega(Z,z_0)$ *is connected, then* $(k,\zeta) = (k,\zeta')$ *for all* $k : X \to Z$, $\zeta,\zeta' : k\,x_0 = z_0$.

Proof. The goal is a proposition. $\zeta' = \zeta\cdot(\zeta^{-1}\cdot\zeta')$ and $\zeta^{-1}\cdot\zeta'$ is merely refl. □

Lemma 2 (pointing irrelevance, circle)**.** *Let* (Z,z_0) *be pointed with* $\Omega(Z,z_0)$ *abelian (e.g. Z pointed equivalent to a loop space,* `EH.Abelian`*). Then two pointed maps* $K_1 \to_* Z$ *with the same underlying map are equal; the same holds for* $\Omega K_2 \to_* Z$ *(precompose with* e_1*).*

Proof. `evPt` is injective, and $\mathtt{evPt}(k,\zeta') = \mathsf{conj}\,\zeta'\,u = \mathsf{conj}\,(\zeta^{-1}\zeta')\,(\mathsf{conj}\,\zeta\,u) = \mathsf{conj}\,\zeta\,u$ with $u = \mathsf{ap}_k\,g$ (`evPt-conj`, `conj-trans`; conjugation by a loop is trivial in an abelian loop group). This is `CupSquare.conj-triv` in general form. □

Lemma 3 (lifting along a map with set fibres)**.** *Let* X *be 1-connected,* $L : (M,m_0) \to_* (P,p_0)$ *with* $\mathrm{fib}_L(p)$ *a set for all* p*. Then* $c \mapsto c; L$ *is a bijection* $\mathrm{Map}_*(X,M) \to \mathrm{Map}_*(X,P)$.

Proof. For $\psi : X \to_* P$ the family $x \mapsto \mathrm{fib}_L(\psi\,x)$ is a family of sets with a point over x_0 (from L_0 and ψ_0). Its pointed sections are contractible (`lemma4`, $c=2$, $t=0$). A pointed section σ gives $c = (\lambda x.(\sigma x)_1, \mathsf{ap}_{\mathrm{pr}_1}\sigma_0)$ with $c; L = \psi$ (homotopy $(\sigma x)_2$, coherence from σ_0); two lifts c,c' of the same ψ give two pointed sections, equal by contractibility, so $c = c'$. □

3. Classes as deloopings of zero

Fix $Y = K^B_4$, $A = K_2$; $\mathsf{PMO}_Y = \mathrm{Map}_*(\Omega K_2, \Omega K^B_4)$, f_0 its base point.

Lemma 4. PMO_Y *is 1-connected (*$\mathsf{isConn}\,3$*) and 2-truncated (*$\mathsf{isTrunc}\,4$*); each* $\mathsf{Unloop}\,f$ *(*$f : \mathsf{PMO}_Y$*) is a set; M_Y is 2-truncated.*

Proof. PMO_Y is a retract of $\mathrm{Map}_*(K_1, K^B_3)$ (`H3Proof.rt-ptmap` with e_1, e_3), which is equivalent to $\Omega K^B_3 \simeq K^B_2$ (`pointed-equiv`, e_2), 1-connected; connectedness passes to retracts at every level (`conn-rt-n`, new: the library's `conn-rt` is for $\mathsf{isConn}\,2$ only). Truncation: `Levels.tPMO` ($c=1$, $t=2$). $\mathsf{Unloop}\,f$ is a set by `Levels[1,2].cor10`: K_2 is $\mathsf{isConn}\,3$ and K^B_4 is $\mathsf{isTrunc}\,6 = \mathsf{isTrunc}(2{+}2{+}2)$. M_Y: `cor5-map` as in `CupSquare.tM4`. □

Proposition 5. $\pi : \mathsf{Unloop}\,f_0 \to [K_2,Y]_*$, $(F,w) \mapsto [F]$*, is a bijection.*

Proof. Surjective: for a class $[F]$ (`HC` induction into propositions), PMO_Y is connected, so merely $w : f_0 = \mathsf{loop}F$. Injective: if $[F] = [F']$ then merely $e : F = F'$ (`effective`); the goal $(F,w) = (F',w')$ is a proposition ($\mathsf{Unloop}\, f_0$ is a set), and $(F,w) = (F', w \cdot \mathsf{ap}_{\mathsf{loop}} e)$; two elements of $f_0 = \mathsf{loop}F'$ are merely equal because this type is connected (PMO_Y is 1-connected, `conn-id`). Conclude with `Bijection.Bij`. □

4. Theorem A: every class is a diagonal

Here $X = K_2$ and Y is any pointed type with $\mathsf{isTrunc}\, 6\, Y$. Everything used follows: PMO and M are $\mathsf{isTrunc}\, 4$; each $\mathsf{Unloop}\, f$ and each $(b : Y) \times C\, a\, b$ is a set (`Levels[1,2]`: `cor10`, `cell`); `cor11` applies to $K_2 \to_* \mathsf{PMO}$; $\Omega(\mathsf{PMO}, f_0)$ is a loop space. Proposition 10 and Lemma 7 hold for any connected A in place of K_2. Write $L = \mathsf{loop} : M \to \mathsf{PMO}$ and $\lambda_0 : L\, 0 = f_0$ (`loop-const`); $\hat{L} = (L, \lambda_0) : M \to_* \mathsf{PMO}$.

Definition 6 (diagonal)**.** For $c : K_2 \to_* M$ with pointing $c_0 : c\, a_0 = 0$: $\Delta c = (\lambda x.\, (c\, x)_1\, x,\ (c\, a_0)_2) : K_2 \to_* Y$. Thus $\kappa \equiv \Delta c_2$.

The invariant. We write out the section of Lemma 9 (its library form goes through `rt-base`, i.e. `Torsor.eqind`, which has no computation rule, so it is not used):

$$s(F,w)\, a = \big(Fa,\ \mathsf{Kf} F\, a,\ \lambda p.\, \mathsf{cell}_{F,w}(a,p)\big), \qquad \mathsf{cell}_{F,w}(a,p) = J(w) : f_0 = D\, a\, (Fa)\, (\mathsf{Kf} F\, a)\, p,$$

$\mathsf{cell}_{F,w}(a_0, \mathsf{refl}) \equiv w$, with the retraction $\mathsf{ret}\, \rho = ((\lambda a.(\rho a)_1,\ H\, a_0\, \mathsf{refl}),\ (\rho a_0)_{2,2} \mathsf{refl} \cdot \mathsf{ap}_D(k)^{-1})$, $H = \lambda a.(\rho a)_{2,1}$, k the a_0-component of `kf-ev` H; $\mathsf{ret}(s\, u) = u$ because `kf-ev` at a J-generated H is refl (J, then $\mathsf{funext}(\lambda_.\mathsf{refl}) = \mathsf{refl}$). So s is injective. Evaluation at a_0 is injective on Rel: each $(b : Y) \times C\, a\, b$ is a set (`Levels.cell`, $t = 2$) and K_2 is connected (`conn-elim` for the propositions $\rho\, a = \rho'\, a$). The proof of `Levels.cell0` transforms $(b : Y) \times C\, a_0\, b$ injectively into pointed sections: $T = \mathsf{tr}_2 \circ s_3 \circ s_2 \circ \mathsf{tr}_1 \circ s_1$, where s_1, s_2, s_3 have retractions and $\mathsf{tr}_1, \mathsf{tr}_2$ are the transports to the centres of the singletons $\mathsf{Sing} = (b : Y) \times (b = b_0)$ and $\mathsf{Sing2} = (h : \mathsf{PMO}) \times (f_0 = h)$; the target is $\mathsf{PtSec} = \mathsf{PtSec}(\Omega K_2, \mathsf{refl}, p \mapsto (f_0 = D\, a_0\, b_0\, f_{0,1}\, p), \ldots)$. Let T' be T followed by forgetting the pointing; T' is injective by Lemma 2: since $D\, a_0\, b_0\, f_{0,1}\, p$ is f_0 by definition, the target is $\mathrm{Map}_*(\Omega K_2, (f_0 = f_0, *))$ with $* = \mathsf{refl} \cdot E\, f_0$ (not refl); loops at $*$ commute by `EH.Abelian` applied to the re-pointing equivalence $(\Omega\mathsf{PMO}, \mathsf{refl}) \simeq_* (f_0 = f_0, *)$, $q \mapsto q \cdot *$. Explicitly $T'(b,h,k)(p) = \bar{k}(p) \cdot \mathsf{ap}_{D(-,p)}(\omega^{-1})$ where $\bar{k}$ is k after tr_1 and $\omega = \bar{k}(\mathsf{refl}) \cdot E^{-1}$ (transport in a family of path types, J).

$$\Theta = T' \circ \mathsf{ev}_{a_0} \circ s :\ \mathsf{Unloop}\, f_0 \to \textstyle\prod_{p:\Omega K_2} (f_0 = f_0) \quad \text{is injective.}$$

The action of defects. For $\varphi : \prod_{p:\Omega K_2} (f_0 = f_0)$ let $\varphi \rhd (b,h,k) = (b, h, \lambda p.\, \varphi(p) \cdot k(p))$ on $(b : Y) \times C\, a_0\, b$, and $(\varphi \cdot \theta)(p) = \varphi(p) \cdot \theta(p)$.

Lemma 7 (equivariance)**.** *If* $\varphi(\mathsf{refl}) = \mathsf{refl}$ *then* $T'(\varphi \rhd v) = \varphi \cdot T'(v)$.

Proof. s_1, s_2 commute with the action by definition; tr_1 acts on the k-component by postcomposition, which commutes with precomposition by $\varphi(p)$ (J on the path in Sing). In the formula for T', ω becomes $\varphi(\mathsf{refl}) \cdot \omega = \omega$. □

The defect of a diagonal.

Lemma 8 (Kf of a diagonal)**.** *For* $c : K_2 \to_* M$, $F = \Delta c$, *all* a *and* $p : a = a_0$: $\mathsf{Kf}\, F\, F_0\, a\, p = \mathsf{Kf}\, (ca)_1\, (ca)_2\, a\, p$.

Proof. J on p (the map $c\, a$ varies with a in the motive); at (a_0, refl) both sides are $(c\, a_0)_2$ by definition. □

Let $K_a : \mathsf{Kf}F\,a = \mathsf{Kf}(ca)\,a$ be its function extensionality. For any $m : M$ let $\mu_m(a,p) : \mathsf{loop}\,m = D\,a\,(m\,a)\,(\mathsf{Kf}m\,a)\,p$ be $J(\mathsf{refl})$ (the canonical multiplicativity of $\mathsf{loop}\,m$); then $\mathsf{cell}_{m,w}(a,p) = w \cdot \mu_m(a,p)$ (J; at refl both are w).

Construction 9. For c, $F = \Delta c$, and $p : a = a_0$,

$$R(a,p) = \big(\mathsf{ap}_L(\mathsf{Kf}c\,a\,p)\cdot\lambda_0\big)^{-1}\cdot\mu_{ca}(a,p)\cdot\big(\mathsf{ap}_{D\,a\,(Fa)\,(-)\,p}\,K_a\big)^{-1}\ :\ f_0 = D\,a\,(Fa)\,(\mathsf{Kf}F\,a)\,p,$$

and $w_c = R(a_0,\mathsf{refl})$; so $\Delta' c = (\Delta c, w_c) : \mathsf{Unloop}\,f_0$.

Proposition 10 (the defect of a diagonal)**.** *For every $c : K_2 \to_* M$, $\mathsf{ev}_{a_0}(s(\Delta' c)) = \varphi_c \rhd v_{00}$, where $v_{00} = \mathsf{ev}_{a_0}(s(0,(\mathsf{refl}\cdot\lambda_0)^{-1}))$ does not depend on c and*

$$\varphi_c(p) = \big(\mathsf{ap}_L(\mathsf{Kf}c\,a_0\,p)\cdot\lambda_0\big)^{-1}\cdot\big(\mathsf{ap}_L(c_0)\cdot\lambda_0\big).$$

Hence $\Theta(\Delta' c) = \varphi_c \cdot \Theta(0,(\mathsf{refl}\cdot\lambda_0)^{-1})$ (Lemma 7; $\varphi_c(\mathsf{refl}) = \mathsf{refl}$ by `trans-sym-l`).

Proof. (i) $\mathsf{cell}_{\Delta c,w_c}(a,p) = R(a,p)$ for all (a,p), by J: at (a_0,refl) both are w_c by definition. (ii) So $\mathsf{ev}_{a_0}s(\Delta' c) = (b, \mathsf{Kf}F\,a_0, \lambda p.R(a_0,p))$ with $b = (c\,a_0)_1 a_0$. Transport along K_{a_0} in the h-component cancels the last factor of R: this equals $(b, \mathsf{Kf}m\,a_0, \lambda p.\,(\mathsf{ap}_L(\mathsf{Kf}c\,a_0\,p)\lambda_0)^{-1}\mu_m(a_0,p))$, $m = c\,a_0$. (iii) Factor out $\varphi_c(p)$: the rest is $(b, \mathsf{Kf}m\,a_0, \lambda p.\,w_m\cdot\mu_m(a_0,p)) = \mathsf{ev}_{a_0}s(m,w_m)$ with $w_m = (\mathsf{ap}_L(c_0)\lambda_0)^{-1}$. (iv) J on $c_0 : m = 0$ (with m free): $\mathsf{ev}_{a_0}s(m,(\mathsf{ap}_L c_0\cdot\lambda_0)^{-1}) = v_{00}$. □

Lemma 11. *$\varphi_c(p) = \big((\mathsf{loop}(c;\hat{L}))_1\,p\big)^{-1}$; so $\varphi_c = \varphi_{c'}$ iff $\mathsf{loop}(c;\hat{L}) = \mathsf{loop}(c';\hat{L})$ (as pointed maps, Lemma 2).*

Proof. `kf-ap`: $\mathsf{Kf}c\,a_0\,p = \mathsf{ap}_c\,p\cdot c_0$; ap_L of a composite; `loop-ap`: $\mathsf{loop}(c;\hat{L})\,p = \mathsf{conj}\,(\mathsf{ap}_L c_0\cdot\lambda_0)\,(\mathsf{ap}_{L\circ c}\,p)$. □

Lemma 12. *$c \mapsto \mathsf{loop}(c;\hat{L})$ is a bijection $\mathrm{Map}_*(K_2,M) \to \mathrm{Map}_*(\Omega K_2, \Omega(\mathsf{PMO}, f_0))$.*

Proof. $\mathsf{loop} : \mathrm{Map}_*(K_2,\mathsf{PMO}) \to \mathrm{Map}_*(\Omega K_2, \Omega\mathsf{PMO})$ has contractible fibres (`cor11`, $c = 1$, PMO is $\mathsf{isTrunc}\,4$), and $c \mapsto c;\hat{L}$ is a bijection by Lemma 3 (fibres of L are Unloop's, sets). □

Theorem 13 (A)**.** *$\Delta' : \mathrm{Map}_*(K_2,M) \to \mathsf{Unloop}\,f_0$ is a bijection; with Proposition 5, $c \mapsto [\Delta c]$ is a bijection $\mathrm{Map}_*(K_2, M_{K_4^B}) \to [K_2, K_4^B]_*$.*

Proof. Let $z = \Theta(0,(\mathsf{refl}\lambda_0)^{-1})$. Injective: $\Delta' c = \Delta' c'$ gives $\varphi_c \cdot z = \varphi_{c'}\cdot z$, so $\varphi_c = \varphi_{c'}$ (cancel pointwise), so $c = c'$ (Lemmas 11, 12). Surjective: for $u : \mathsf{Unloop}\,f_0$ let $\psi(p) = \Theta(u)(p)\cdot z(p)^{-1}$; it is pointed ($\Theta(u)(\mathsf{refl})$ and $z(\mathsf{refl})$ are both the base point of PtSec up to the pointings); by Lemma 12 there is c with $\mathsf{loop}(c;\hat{L}) = \psi^{-1}$ (pointwise inverse), so $\varphi_c = \psi$ and $\Theta(\Delta' c) = \Theta(u)$, hence $\Delta' c = u$. □

5. Theorem B: the bilinear classification

For a pointed $G : Y \to_* Y'$ let $G_* : M_Y \to_* M_{Y'}$, $m \mapsto m;G$, pointed by `comp-const-l`. For $c : K_2 \to_* M_Y$ let

$$\mathcal{E}(c) = \mathsf{ev}_{\Omega Y}\big(\mathsf{evPt}(e_1^{-1};\mathsf{loop}\,c;\mathsf{ev}_Y)\big) : K_2 \to_* \Omega^2 Y,$$

the composite of looping, the mapping-loop equivalences and the circle property, bracketed as `c1bar`, $e_1^{-1};(\mathsf{loop}c;\mathsf{ev}_Y)$, so that $\mathcal{E}(c_2)$ is by definition the left side of `cup-norm`: $\mathcal{E}(c_2) = \iota$.

Lemma 14. *$\mathcal{E}$ is injective (for $Y = K_4^B$).*

Proof. loop is injective on $\mathrm{Map}_*(K_2,M_Y)$ (`unique-loop`, M_Y is $\mathsf{isTrunc}\,4$); pre- and postcomposition with pointed equivalences, `evPt` and `ev` are injective. □

Lemma 15 (naturality)**.** *$\mathcal{E}(c;G_*) = \mathcal{E}(c);\mathsf{loop}^2 G$ for pointed $G : Y \to_* Y'$, when $\mathrm{Map}_*(K_2,\Omega^2 Y')$ is a set and $\Omega^3 Y'$ is connected (e.g. $Y' = K_4^B$).*

Proof. By Lemma 1 it suffices to compare underlying maps. $\mathsf{loop}(c;G_*) = \mathsf{loop}c;\mathsf{loop}G_*$ (`loop-comp`); `evPt` is natural: $\mathtt{evPt}(\psi;N) = \mathsf{loop}N(\mathtt{evPt}\,\psi)$ (`evPt-loop`, `loop-comp`). With $N = \mathsf{ev}_Y^{-1};\mathsf{loop}G_*;\mathsf{ev}_{Y'}$ (`ev-unev`) and $\ell = \mathtt{evPt}(e_1^{-1};\mathsf{loop}c;\mathsf{ev}_Y)$ we must show, for each x, $\mathsf{ap}_{\mathrm{ev}_x}(\mathsf{loop}N\,\ell) = \mathsf{loop}^2G\,(\mathsf{ap}_{\mathrm{ev}_x}\ell)$ (`ev-ap` on both sides). As in `cup-norm`: take $\mathrm{ev}_x : \mathrm{Map}_*(K_2,\Omega Y') \to_* \Omega Y'$ (pointed by refl), so both sides are loopings, $\mathsf{loop}(N;\mathrm{ev}_x)\,\ell$ and $\mathsf{loop}(\mathrm{ev}_x;\mathsf{loop}G)\,\ell$ (`loop-comp`). The pointed maps $N;\mathrm{ev}_x$ and $\mathrm{ev}_x;\mathsf{loop}G$ are homotopic (unpointed), by `ev-ap`, `loop-ap`, an "ap of conj" lemma (J), $\mathsf{ap}_{\mathrm{ev}_x}\pi_G = \mathsf{refl}\cdot G_0$ (`happly-funext`) and $\mathsf{ev}^{-1}\circ\mathsf{ev} = \mathrm{id}$ (from injectivity of ev; `unev` is locked). By `LoopConj.LoopHomotopy` their loopings differ by conjugation by a loop of $\Omega Y'$ at refl, i.e. a 2-loop, acting on a 2-loop: trivial by `EH.eh`. □

Lemma 16 (H2)**.** *For abelian A,B, $g \mapsto K(g,1)$ and $g \mapsto K(g,2)$ are bijections* $\mathrm{Hom}(A,B) \to \mathrm{Map}_*(K_1^A,K_1^B)$, $\to \mathrm{Map}_*(K_2^A,K_2^B)$.

Proof. Level 1. $\mathsf{loop}K(g,1) = \mathsf{conj}_0(g) = \mathsf{unband}\circ g\circ\mathsf{band}$ (`loopK 0`, `c0-conj`). Injective: equal maps have equal loopings, so $g\circ\mathsf{band} = g'\circ\mathsf{band}$, and band is surjective. Surjective: for F, $g_F = \mathsf{band}\circ(\mathsf{loop}F)_1\circ\mathsf{unband}$ is a homomorphism (`loop-mult`, `band-hom`); $\mathsf{loop}K(g_F,1) = \mathsf{loop}F$ pointwise (`band-unband`, `unband-band`), pointings in a set; so $K(g_F,1) = F$ (`unique-loop1`, $c=0$). Level 2. $\mathsf{loop}K(g,2) = e_1;K(g,1);e_1^{-1}$ (`loopK 1`); conjugation by e_1 is a bijection $\mathrm{Map}_*(K_1^A,K_1^B) \to \mathsf{PMO}$; loop is injective (`unique-loop`, $c=1$) and every PMO element is a looping (`cor11`). □

Lemma 17 (Ω^2 of $K(g,4)$)**.** *Let $j_B = \mathsf{loop}e_3;e_2 : \Omega^2K_4^B \to_* K_2^B$ (so $\iota = j_{\mathbb{Z}}^{-1}$). Then $\iota;\mathsf{loop}^2K(g,4) = K(g,2);j_B^{-1}$.*

Proof. $\mathrm{Map}_*(K_2,\Omega^2K_4^B)$ is a set and $\Omega^3K_4^B$ is connected, so by Lemma 1 it suffices to compare underlying maps after postcomposing with j_B: `loopK` at 3 and 2, `loop-comp`, `pt-inv-r` and `Puppe.loop-id` pointwise. ($j_{\mathbb{Z}}$ is by definition `iota-eq.1`.) □

Theorem 18 (B)**.** *$g \mapsto c_2;K(g,4)_*$ is a bijection* $\mathrm{Hom}(\mathbb{Z},B) \to \mathrm{Map}_*(K_2,M_{K_4^B})$.

Proof. $\mathcal{E}(c_2;K(g,4)_*) = \iota;\mathsf{loop}^2K(g,4) = K(g,2);j_B^{-1}$ (Lemmas 15, 17, `cup-norm`). Injective: by Lemma 16 and injectivity of $-;j_B^{-1}$. Surjective: for c, $\mathcal{E}(c);j_B = K(g,2)$ for some g (Lemma 16), so $\mathcal{E}(c) = \mathcal{E}(c_2;K(g,4)_*)$ and $c = c_2;K(g,4)_*$ (Lemma 14). □

6. (H4), its inverse, and the coefficients

Theorem 19. $\mathsf{h4} : \mathsf{H4}\ \kappa$.

Proof. By definition $\Delta(c_2;G_*) = (\lambda x.\,G((c_2x)x),\ \mathsf{ap}_G(c_2a_0)_2\cdot G_0) \equiv \kappa;G$. Hence $\Phi_B = [-]\circ\Delta\circ(g \mapsto c_2;K(g,4)_*)$, a composite of bijections of sets (Theorems 18, 13); `Bijection.Bij` turns it into `isEquiv`, i.e. an inverse Ψ_B with both equations. □

(a) The inverse. Ψ_B is, explicitly, the composite of the inverses: a class $[F]$ is represented by (F,w) (Prop. 5); c is the bipointed map with $\mathsf{loop}(c;\hat{L}) = (\Theta(F,w)\cdot z^{-1})^{-1}$ (Lemma 12: delooping by `cor11`, then lifting); and g is $K(-,2)^{-1}(\mathcal{E}(c);j_B)$, that is, with $G = e_1^{-1};\mathsf{loop}(\mathcal{E}(c);j_B);e_1 : K_1 \to_* K_1^B$, $g = \mathsf{band}\circ(\mathsf{loop}G)_1\circ\mathsf{unband}$ (Lemma 16). Both equations $\Psi_B\Phi_B = \mathrm{id}$, $\Phi_B\Psi_B = \mathrm{id}$ are the `Bij` equations.
(b) The generator is the constructed κ. It enters exactly twice: $\kappa \equiv \Delta c_2$ (definition), and $\mathcal{E}(c_2) = \iota$, which is `cup-norm` (itself from `iota`, `c1-ev`, `c2-loop`, `ev-ap` through $\mathcal{E}$'s definition).
(c) Coefficients. B is arbitrary (including $\mathbb{Z}/2$); naturality: for $h : B \to B'$, $[\Phi_B(g);K(h,4)] = \Phi_{B'}(h\circ g)$ by `KFun` ($K(h\circ g,4) = K(g,4);K(h,4)$) and associativity of composition of classes. Φ_B is the map of `KZ2Cohomology.Phi` (it uses `KMap[Zab,B,g].K four`).

Remark (the trap). The invariant Θ is not quadratic in disguise: for $B = \mathbb{Z}/2$ and $g = 1$, Theorem B gives $c_2; K(1,4)_* \neq 0$, and Theorem A gives $[\kappa; K(1,4)] \neq 0$; a cross-term invariant would see $2 \cdot 1 = 0$.

7. Dependency audit (requirement d)

No Gysin or transgression sequence is used. The role of the path fibration $K(\mathbb{Z},1) \to * \to K(\mathbb{Z},2)$ is played by DW1's Lemma 9: the cell of a delooping is defined over the based path space $(a : K_2) \times (a = a_0)$ by J, and Proposition 10(i) is a J over it. No pushout, no suspension, no join, no truncation beyond `HC` (set truncation, from `Quotient`).

Levels. K_2: isConn 3 (1-connected); K_4^B: isConn 5, isTrunc 6. Unloop f: `Levels[c=1,t=2]`, sets (isTrunc$(2+2+2)$). PMO: isTrunc 4 (`tPMO`), isConn 3 (Lemma 4). M: isTrunc 4 (`cor5-map 4 2`). `cor11` at $c = 1$ for $K_2 \to_* M$ and $K_2 \to_*$ PMO (isTrunc 4); `unique-loop` at $c = 1$; `unique-loop1` at $c = 0$ (K_1^B is isTrunc 3); `lemma4` with $c = 2, t = 0$ (sets over a 1-connected base) in Lemma 3. $\mathrm{Map}_*(K_2, \Omega^2 K_4^B)$ is a set (`cor5-map 2 2`), $\Omega^3 K_4^B$ is connected (`conn-id` three times from isConn 5).

Library lemmas used: (I1)–(I8). **New lemmas:** `conn-rt-n`; the re-pointing pointed equivalence for `EH.Abelian`; the explicit section/retraction of Lemma 9; transport in families of path types (the action; $\mathsf{tr}_1, \mathsf{tr}_2$); lifts as pointed sections (Lemma 3); "ap of conj"; $\mathsf{ev}^{-1} \circ \mathsf{ev} = \mathrm{id}$; post-composition on classes for (c); Lemmas 1, 2, 3, 4, Prop. 5; injectivity of Θ; Lemma 7, 8, the μ/cell lemma, Construction 9, Prop. 10, Lemmas 11, 12, Theorem A; Lemmas 14, 15, 16, 17, Theorem B; Theorem 19. No library file is modified.

8. Correspondence with the formal proof

Status: checked (68 modules, 0 failed; 2771 declarations, 1.1 s; `h4.opens` generated from `-opened`, not yet tight).

- `H4Classes`: `conn-rt-n`, Lemma 1 `pi-set`; `ClassesAt[B]`: Lemma 4 (`pmo-conn`, `unloop-set`), Prop. 5 (`cls`, `cls-inj`, `cls-surj`, `cls-equiv`).
- `H4Diagonal`: `Section` (the Lemma 9 section `sec`, retraction `ret`, `sec-inj`); `RelEval.ev-inj`; `ToCentre`; `Defect` (`act`, `pre-formula`, Lemma 7 `inv-act`); `Diag` (Lemma 8 `kf-diag`, `mu`, `cell-mu`, Construction 9 `R`, `wc`, `dprime`, Prop. 10 `cell-R`, `ev-diag`, `inv-diag`); `PICircle` (Lemma 2); `repoint`; `DiagAt[B]`: `theta-inj`, Lemma 3 (`Lift`, `lift-unique`), Lemma 11 `phi-loop`, Theorem A (`diag-inj`, `undiag`, `diag-surj`).
- `H4Bilinear`: `H2` (Lemma 16: `k1-inj`, `k1-surj`, `k2-inj`, `k2-surj`); `EDef` ($\mathcal{E}$); `ENat` (Lemma 15 pointwise: `nat-pt`, `hh`, `e-nat`); `BilAt[B]`: `theta-inj` (Lemma 14), Lemma 17 `j-pt`, `key`, Theorem B (`c2g-inj`, `c2g-surj`).
- `H4Proof`: `H4At[B]` (`inj`, `surj`, `equiv`), `h4`.